— Solution notes —

COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 6

5 Computation Theory (amp12)

Register machines; decidable sets

(*a*) Explain what it means for a partial function $f : \mathbb{N}^n \rightharpoonup \mathbb{N}$ to be register machine *computable* and for a set of numbers $S \subseteq \mathbb{N}$ to be register machine *decidable*. [5 marks]

Answer: $f \in \mathbb{N}^n \rightharpoonup \mathbb{N}$ is register machine computable if there is a register machine M with at least $n+1$ registers $\mathtt{R}_0$, $\mathtt{R}_1$, ..., $\mathtt{R}_n$ (and maybe more) such that for all $(x_1, \ldots, x_n) \in \mathbb{N}^n$ and all $y \in \mathbb{N}$, the computation of M starting with $\mathtt{R}_0 = 0$, $\mathtt{R}_1 = x_1$, ..., $\mathtt{R}_n = x_n$ and all other registers set to 0, halts with $\mathtt{R}_0 = y$ if and only if $f(x_1, \ldots, x_n) = y$.

$S \subseteq \mathbb{N}$ is register machine decidable if its characteristic function $\chi_S : \mathbb{N} \to \mathbb{N}$

$$\chi_S(n) \triangleq \begin{cases} 1 & \text{if } n \in S \\ 0 & \text{if } n \notin S \end{cases}$$

is register machine computable.

(*b*) A set of numbers $S \subseteq \mathbb{N}$ is register machine *enumerable* if either S is empty, or $S = \{f(n) \mid n \in \mathbb{N}\}$ for some total function $f : \mathbb{N} \to \mathbb{N}$ which is register machine computable.

(*i*) Show that if S is register machine decidable, then it is register machine enumerable. [5 marks]

Answer: If $S = \emptyset$, then it is RM enumerable, so suppose S is non-empty, say $n_0 \in S$. Since S is RM decidable there is a RM M computing χ_S. Modify M's program so that it first copies the contents of $\mathtt{R}_1$ to a fresh (for M's program) register $\mathtt{X}$, runs M's program to compute $\chi_S(\mathtt{R}_1)$ in $\mathtt{R}_0$ (guaranteed to halt, because χ_S is total), then copies $\mathtt{X}$ to $\mathtt{R}_0$ if $\mathtt{R}_0 = 1$, else copies n_0 to $\mathtt{X}$, and then halts. Thus the modified program computes the function $f : \mathbb{N} \to \mathbb{N}$

$$f(n) \triangleq \begin{cases} n & \text{if } n \in S \\ n_0 & \text{if } n \notin S \end{cases}$$

whose image is S. Thus S is RM enumerable.

(*ii*) Show that if both S and its complement $\overline{S} \triangleq \{n \in \mathbb{N} \mid n \notin S\}$ are register machine enumerable, then S is register machine decidable. [5 marks]

Answer: If either of S or $\overline{S}$ are empty, then χ_S is a constant function and hence RM computable. So we can assume there are RM computable functions $f_1 : \mathbb{N} \to \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ whose images are S and $\overline{S}$ respectively.

Because f_1 and f_0 are total functions, from the RMs computing them we can construct a RM M that, on inputting n, computes the sequence

$$f_1(0), f_0(0), f_1(1), f_0(1), f_1(2), f_0(2), \ldots$$

until the value n appears in the sequence (which is guaranteed to happen because between them f_1 and f_0 enumerate the whole of $\mathbb{N}$), $n = f_i(k)$ say, and then halts with value i in $\mathtt{R}_0$. Thus M computes χ_S and S is RM decidable.

(iii) Give an example of a set of numbers that is register machine enumerable, but not register machine decidable. (Any standard results about computable functions that you use should be carefully stated.) [5 marks]

Answer: Using the *Undecidability of the Halting Problem* [statement] one has that the set

$$S \triangleq \{\langle \ulcorner P \urcorner, \ulcorner [a_1, \ldots, a_n] \urcorner \rangle \mid P \text{ is a RM program that halts when started with } \mathtt{R}_i = a_i \ (i = 1, \ldots, n) \text{ and all other registers zeroed} \}$$

is not RM decidable (where $(m, n) \mapsto \langle m, n \rangle$, $\ell \mapsto \ulcorner \ell \urcorner$ and $P \mapsto \ulcorner P \urcorner$ are effective bijective codings of pairs, lists and RM programs).

However, S is RM enumerable because of the existence of a *Universal RM Program* [statement]: modify that to decode the argument in $\mathtt{R}_0$ to $\langle \langle \ulcorner P \urcorner, \ulcorner [a_1, \ldots, a_n] \urcorner \rangle, t \rangle$ and then run t steps of the universal RM simulating the computation of P started with arguments $a_1, \ldots, a_n$, returning $\langle \ulcorner P \urcorner, \ulcorner [a_1, \ldots, a_n] \urcorner \rangle$ if it does halt after t steps and returning some fixed element of S if it does not. Thus this modified RM computes a total function enumerating S.
